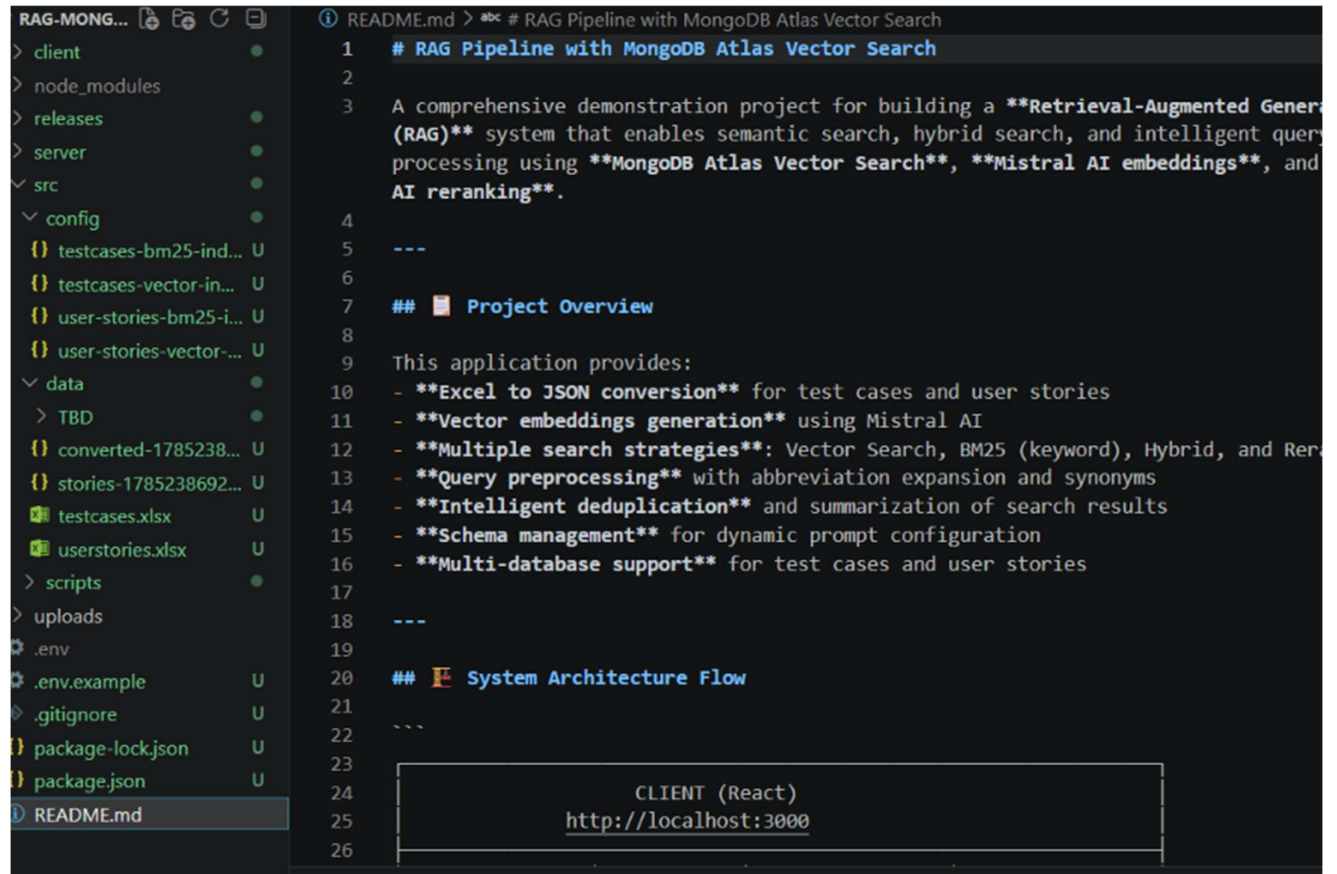


Create a readme.md file to understand the flow, impacted files and API's for ingestion.

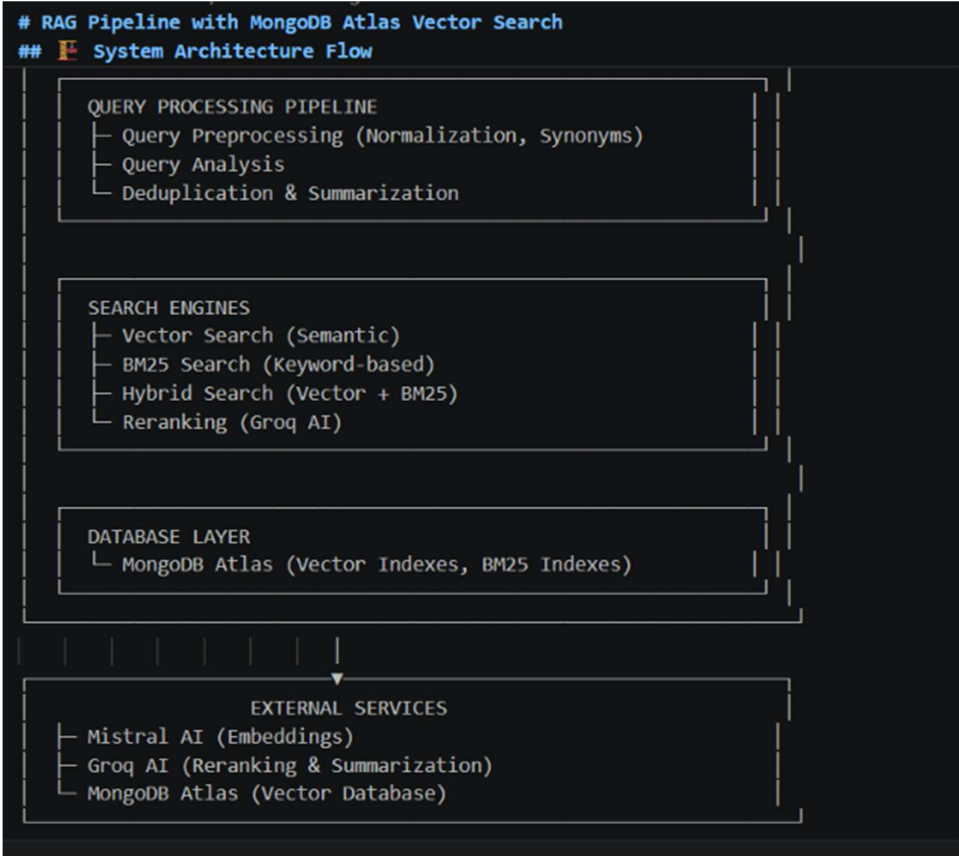
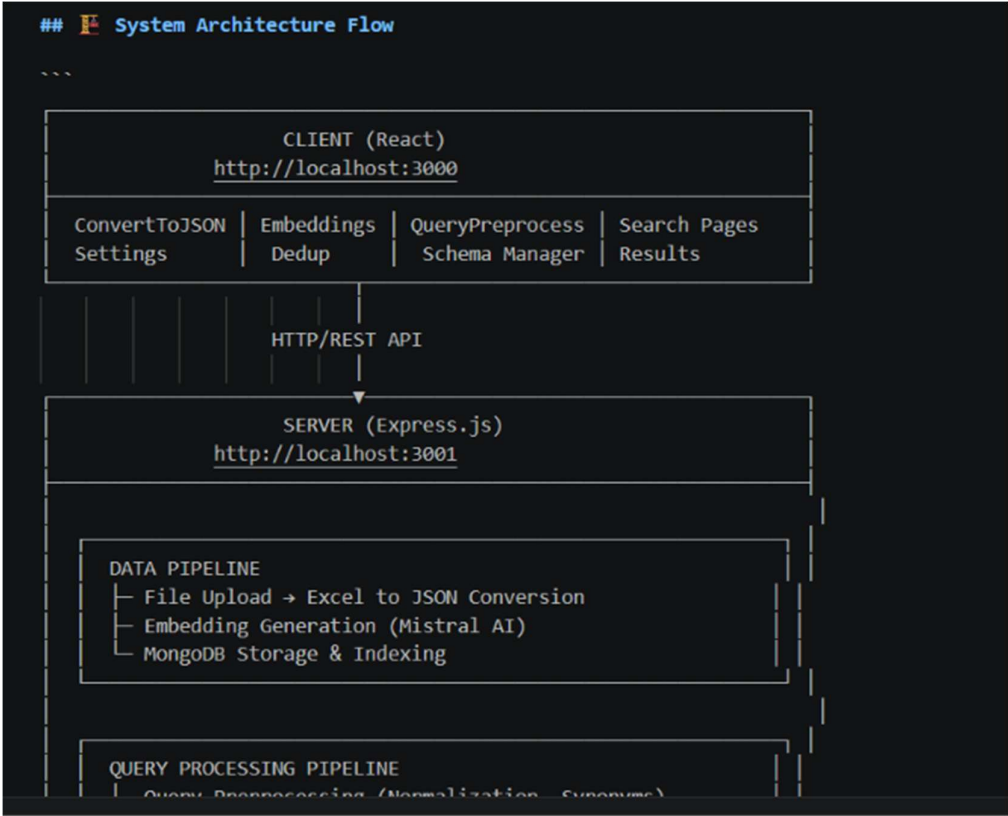
Prompt given:

Analyze the project and create the readme.md file to understand the flow, impacted files, backend frontend integration details and API's



The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders like 'client', 'node_modules', 'releases', 'server', 'src', 'config', 'data', 'scripts', and 'uploads'. The code editor shows the content of 'README.md'.

```
1 # RAG Pipeline with MongoDB Atlas Vector Search
2
3 A comprehensive demonstration project for building a Retrieval-Augmented Generation (RAG) system that enables semantic search, hybrid search, and intelligent query processing using MongoDB Atlas Vector Search, Mistral AI embeddings, and AI reranking.
4
5 ---
6
7 ## 📋 Project Overview
8
9 This application provides:
10 - Excel to JSON conversion for test cases and user stories
11 - Vector embeddings generation using Mistral AI
12 - Multiple search strategies: Vector Search, BM25 (keyword), Hybrid, and Reranking
13 - Query preprocessing with abbreviation expansion and synonyms
14 - Intelligent deduplication and summarization of search results
15 - Schema management for dynamic prompt configuration
16 - Multi-database support for test cases and user stories
17
18 ---
19
20 ## 🏗️ System Architecture Flow
21
22 ---
23
24 CLIENT (React)
25 http://localhost:3000
26
```



```

## 🗄 Database Schema

### Test Cases Database
- Database: `db_testcases`
- Collection: `testcases`
- Vector Index: `vector_index_test_cases`
- BM25 Index: `bm25_search`

### User Stories Database
- Database: `db_testcases` (same)
- Collection: `userstories`
- Vector Index: `vector_index_user_story`
- BM25 Index: `bm25_user_stories`

---

## 📊 Data Flow Diagrams

### 1. Data Ingestion Pipeline
...
Excel File
  ↓
/api/upload-excel
  ↓
Excel to JSON Conversion
  ↓
ConvertToJsonComponent

```

📁 Included Sections:

1. Project Overview - What the application does
2. System Architecture Flow - Visual ASCII diagram showing client → server → database → external services
3. Database Schema - Structure for test cases and user stories
4. Data Flow Diagrams - 3 detailed flows (ingestion, search, query processing)
5. API Endpoints - Complete 20+ endpoint reference with methods, paths, and purposes
6. Backend Architecture - All 5 core module groups with file descriptions
7. Frontend Architecture - 4 component categories with purposes
8. Backend-Frontend Integration Points - 4 key integration flows with visual paths
9. Impacted Files by Feature - Mapping of files for each major feature (7 features covered)
10. Environment Configuration - All required `.env` variables
11. Key Features - 5 major capabilities highlighted
12. Setup & Running - Installation and execution instructions
13. Performance Considerations - Optimization tips
14. Security Notes - Security best practices
15. Project Structure - Directory tree organization
16. Troubleshooting Guide - Common issues and solutions